Stock Optimization Modeling

Homework Team 2
Carlos Alonzo-Montufar, Ali Awan,
Delaney Black, Liam Cawley, and Priya
Chilukuri

Friday, March 13, 2026

Table of Contents

Overview **1**
Methodology **1**
Data Used *1*
Model Development & Forecasting *1*
Analysis **1**
Results & Recommendations **2**
Conclusion **3**

STOCK OPTIMIZATION MODELING

Overview

The Commercial Banking Corporation (the "Bank") aims to minimize portfolio risk while achieving a target daily return of at least 0.18% using five prominent technology stocks: NVIDIA (NVDA), Google (GOOG), Broadcom (AVGO), Apple (AAPL), and Meta (META). Our team collected daily stock prices and returns for these securities from January 2, 2025, to December 31, 2025. A Gurobi quadratic optimization model was used to determine the optimal asset allocation that minimizes the risk for a daily return under the specified return constraint. To meet the 0.18% daily return objective with the lowest possible risk, we recommend a heavy allocation in GOOG at approximately 69.10%, followed by AAPL at 19.70% and META at 8.80%. AVGO should represent 2.50% of the portfolio, while NVDA should be excluded or kept at a negligible weight to maintain the risk-reward balance.

Methodology

This section outlines our data acquisition and model development process.

Data Used

We first collected the percent daily returns from January 2, 2025, to December 31, 2025, for the Nvidia, Google, Broadcom, Apple, and Meta stocks. The scraped dataset included 1820 observations, and the end-of-day stock value for each stock. The data was scraped from Yahoo Finance.

Model Development & Forecasting

To determine the percentage of the portfolio to allocate to each stock, we used an optimization model that minimized portfolio risk. We added constraints to ensure that the sum of proportions for the stocks equaled 1 and that the daily return was at least 0.18%. To assess different portfolio strategies for different levels of risk, we calculated the "efficient frontier," which provides insight into portfolios that offer the lowest risk for an expected return or the highest expected return for a given level of risk.

Analysis

The following analysis illustrates how we obtained the optimal portfolios for low-, medium-, and high-risk appetites.

To achieve a daily return of 0.18% with minimal risk, our team used Gurobi Optimizer to identify the optimal portfolio while maintaining the same resource constraints. We then created an efficient frontier to visualize the relationship between potential returns and associated risk, which helped identify optimal portfolios for our varying risk appetites.

Results & Recommendations

After running the optimization to minimize risk, we got the following efficient frontier graph, displayed in Figure 1.

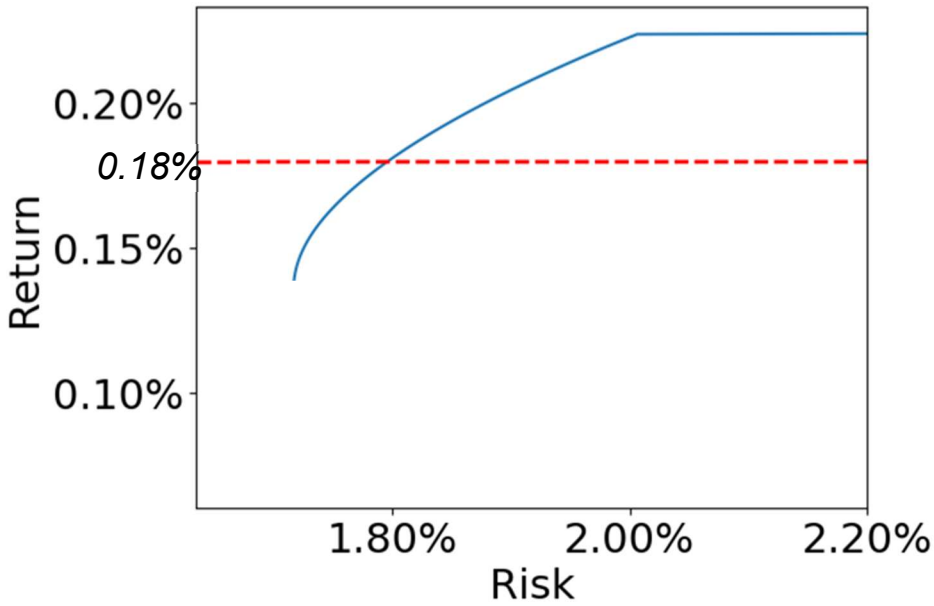

Figure 1: Efficient Frontier Graph

The minimum required return given by the Bank is 0.18%, as shown by the horizontal dotted line. The global minimum risk is about 1.70%, which is associated with an expected daily return of about 0.14%. The efficient frontier also shows the maximum feasible return for the whole portfolio, which is about 0.22%, beyond which new stocks would have to be included to achieve a greater expected return.

The optimization gave us multiple stock allocations of varying risk levels. The following low-risk allocation is expected to return 0.18% daily with a risk of 1.80%: 19.70% of AAPL, 8.80% of META, 2.50% of AVGO, 69.10% of GOOG, and approximately 0% of NVDA. If one's risk appetite allows, this medium-risk allocation is expected to return 0.20% daily with a risk of 1.89%: 10.90% of AAPL, 2.90% of META, 5.30% of AVGO, 80.80% of GOOG, and 0% of NVDA. Finally, this high-risk stock allocation is expected to return 0.22% daily with a risk of 2.01%: 92.39% of AVGO and 7.60% of GOOG.

NVDA does not appear in any of the optimized stock portfolios. We noticed that NVDA had the second-highest volatility overall while being only third in terms of average returns. It's possible that NVDA is too volatile for the optimization to include it, and that volatility may be at least partially driven by the very recent demand for computer chips to train and run AI.

If the Bank wants to minimize risk, we recommend following the lowest-risk allocations. If the Bank wants to maximize returns, we recommend following either the medium or high-risk allocations, depending on the Bank's risk appetite.

Conclusion

Given the Bank's minimum return of 0.18%, the viable portfolio risk range is from 1.80% to 2.01%, and the viable expected daily returns range from 0.18% to 0.22%. After reviewing 2025 stock performance data and running an optimization model to minimize risk, we calculated stock portfolio allocations based on the investor's risk tolerance. Out of the low, medium, and high-risk allocations, all of them contained at least some of GOOG and AVGO. In general, GOOG and AVGO seem to be high-returning stocks with manageable volatility, which may explain why the optimizer includes them in even the low-risk portfolio. In the future, if the Bank wants to target expected daily returns higher than 0.22%, it will have to bring in other potential stocks and likely increase its risk tolerance as well.